An explicit two-component mollifier for zeros of ξ' on the critical line

27 October 2026 — Draft for expert audit

Abstract

We give a proof draft for a lower bound of 88.45% for the proportion of zeros of the derivative of the Riemann xi-function on the critical line, counted with multiplicity. The construction combines the argument-variation method used by Wu with an unrestricted Conrey–Feng mollifier of fixed length $T^{4/7-10^{-6}}$. Only the convolution $\mu * \Lambda * \Lambda$ is added to the Conrey component. All polynomials are specified by exact finite decimals. We give the specialized leading-pole functional and an interval certificate for its value. The analytic transplantation is presented for specialist verification; the finite certificate does not verify that transplantation. No assertion about simple zeros or publication priority is made.

1 Statement and explicit data

Put

$$\xi(s) = H(s)\zeta(s), \qquad H(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2).$$

Let $N(T)$ count the nontrivial zeros of ζ with $0 < \operatorname{Im}\rho \le T$, with multiplicity, and let $N_{\xi',c}(T)$ count zeros of ξ' in this height range on $\operatorname{Re} s = 1/2$, also with multiplicity. The standard count for ξ' is $N(T) + O(\log T)$; thus the denominator below also gives its asymptotic proportion of all zeros.

Theorem 1 (Claim submitted for verification)**.** *The explicit construction below gives*

$$\liminf_{T\to\infty} \frac{N_{\xi',c}(T)}{N(T)} > 0.8845.$$

This updates the multiplicity-counted quantity in Wu's Section 3 [1], not a simple-zero proportion. In particular, it is a different quantity from the 0.86864 simple-and-on-line bound stated in [3, Remark 7.1]. We fix, without further optimization,

$$R = 1.04440033, \qquad \theta = \frac{4}{7} - 10^{-6}, \qquad L = \log T, \qquad a = \frac{1}{2} - \frac{R}{L}, \quad y = T^{\theta}. \tag{1}$$

The operator and two profiles are

$$\begin{aligned} q(x) &= 1 - 2.27043254x + 0.304358115x^2 + 0.16793368x^3 \\ &\quad + 4.34152016x^4 - 5.3105844x^5 + 1.7701948x^6, \end{aligned} \tag{2}$$

$$P_1(x) = -0.30528514x + 2.50038954x^2 - 1.12177152x^3 - 0.07333288x^4, \tag{3}$$

$$P_2(x) = 1.89779072x + 0.56594804x^2 + 0.07023055x^3. \tag{4}$$

Every displayed decimal in (1)–(4) is an exact rational number. In particular $q(0) = P_1(1) = 1$ and $P_1(0) = P_2(0) = 0$. With $x_n = \log(y/n)/\log y$, define

$$\Psi(s) = \sum_{n\le y} \frac{1}{n^{s+R/L}} \left\{ \mu(n)P_1(x_n) + \frac{(\mu*\Lambda*\Lambda)(n)}{(\log y)^2} P_2(x_n) \right\}. \tag{5}$$

Here $*$ denotes Dirichlet convolution and Λ is the von Mangoldt function. There is *no* factor $\mu^2(n)$ in the second component and no extra factorial. Non-squarefree terms are retained. Write

$$U(s) = q(-L^{-1}\partial_s)\zeta(s). \tag{6}$$

The operator acts only on ζ, never on Ψ.

2 The counting interface

The following formulation separates the exact counting function from its asymptotic differential operator. It is the argument principle and Littlewood reduction of [1, Section 3, (18)–(23)], with the right boundary normalization checked for (5).

Lemma 2. *For the fixed data above, suppose*

$$\frac{1}{T}\int_T^{2T} |U(a+\mathrm{i}t)\Psi(a+\mathrm{i}t)|^2\,\mathrm{d}t = M + o(1), \tag{7}$$

and the same mollified second moments for each $L^{-k}\zeta^{(k)}$, $0 \le k \le 6$, are bounded. Then

$$\liminf \frac{N_{\xi',c}(T)}{N(T)} \ge 1 - \frac{\log M}{R}. \tag{8}$$

Proof. Set $r(u) = q(1/2-u)$. Exact polynomial arithmetic gives

$$r(u) = 0.09491351 + 0.997010185u + 2.089866u^2 - 2.29671034u^4 + 1.7701948u^6. \tag{9}$$

If $r(u) = \sum r_j u^j$, use the exact auxiliary function

$$\eta(s) = \sum_{j=0}^{6} r_j L^{-j}\xi^{(j)}(s), \qquad V(s) = \eta(s)/H(s). \tag{10}$$

On the critical line, $\xi^{(2j)}$ is real and $\xi^{(2j+1)}$ is purely imaginary. Thus

$$\operatorname{Im}\eta(1/2+\mathrm{i}t) = \frac{0.997010185}{L}\operatorname{Im}\xi'(1/2+\mathrm{i}t).$$

The nonzero coefficient is essential. The phase-counting argument of Wu therefore applies directly, without identifying his polynomial parameters with an unverified exact auxiliary function. Argument variation is taken upward on the line, passing boundary zeros on their east side. Each net advance of π detects a zero of ξ'; zeros common to η and the line are included with multiplicity.

On the fixed window $T \le t \le 2T$, Stirling's formula gives, uniformly in a fixed strip around $1/2 \le \operatorname{Re} s \le 3$,

$$\frac{H^{(j)}(s)}{H(s)L^j} = 2^{-j} + O(L^{-1}).$$

Since $r(1/2+z) = q(-z)$, it follows that

$$V - U = \sum_{k=0}^{6} O(L^{-1})L^{-k}\zeta^{(k)}, \qquad \frac{1}{T}\int_T^{2T} |(V-U)\Psi|^2\,\mathrm{d}t = O(L^{-2}).$$

The coefficients in the first error may depend on t but are uniformly bounded as stated. Consequently the moment of $V\Psi$ is $M + o(1)$.

At $\operatorname{Re} s = 3$, absolute convergence and the profile endpoints show

$$\Psi_1(s) = \zeta(s)^{-1} + O(L^{-1}), \qquad \Psi_2(s) = O(L^{-2}), \qquad V(s) = \zeta(s) + O(L^{-1}).$$

For the first estimate, bound the profile error by $O(\log n/\log y)$ and also include the factor $n^{-R/L}$ and the truncation tail. For the second, use $\sum_n |\mu * \Lambda * \Lambda|(n)n^{-3} < \infty$. Hence $V\Psi = 1 + O(L^{-1})$ on the right boundary. The logarithmic integral there is $O(T/L)$; it is not necessary that $\Psi \to 1$.

Let n_V^+ count zeros of V in the right rectangle $1/2 \le \operatorname{Re} s \le 3$, $T \le \operatorname{Im} s \le 2T$, using the boundary convention of Wu. Its argument principle and $\Delta \arg H = \pi(N(2T) - N(T)) + O(\log T)$ give

$$N_{\xi',c}(2T) - N_{\xi',c}(T) \ge N(2T) - N(T) - 2n_V^+ + O(\log T). \tag{11}$$

An equivalent safe convention overcounts boundary zeros on the right. Heights hitting zeros are treated by limiting admissible heights. Polynomial growth of the fixed-degree functions, together with their nonzero right-boundary normalization, gives the horizontal argument integrals $O(\log T)$ by the usual Jensen argument [1, Section 2, (6)–(8)]. There are no poles of H or ζ in these high positive-height rectangles.

Every zero counted by n_V^+ is a zero of $V\Psi$ and is at horizontal distance at least R/L from $\operatorname{Re} s = a$. Littlewood's formula then gives

$$2\pi\frac{R}{L}n_V^+ \le \int_T^{2T} \log|V(a+\mathrm{i}t)\Psi(a+\mathrm{i}t)|\,\mathrm{d}t + O(T/L) + O(\log T).$$

Apply Jensen's inequality here, *before* replacing V by U. No comparison of their negative logarithmic tails is used. Inserting the second-moment estimate and (11), and using $N(2T)-N(T) \sim TL/(2\pi)$, yields the local version of (8). Add finitely many dyadic windows $[T/2^J, T]$, with their own fixed L and mollifier. First let $T\to\infty$, then $J\to\infty$; the omitted count is a fraction $2^{-J}+o(1)$. No estimate uniform in J is required. □

3 Shifted moment and the specialized kernel

We use the third coefficient case of Pratt–Robles–Zaharescu–Zeindler [2, Theorem 4.1, p. 18]. For fixed smooth height weight Φ, small complex shifts $\alpha,\beta = O(L^{-1})$, and the two coefficient sequences obtained by expanding (5), the theorem gives

$$\begin{aligned}&\int \zeta(\tfrac12+\alpha+\mathrm{i}t)\zeta(\tfrac12+\beta-\mathrm{i}t)\Psi(a+\mathrm{i}t)\overline{\Psi(a+\mathrm{i}t)}\Phi(t/T)\,\mathrm{d}t\\ &\quad=\sum_{d,e\le y}\frac{a_d a_e}{[d,e]}\frac{(d,e)^{\alpha+\beta}}{d^\alpha e^\beta}\int\left\{\zeta(1+\alpha+\beta)+\zeta(1-\alpha-\beta)\left(\frac{2\pi de}{t(d,e)^2}\right)^{\alpha+\beta}\right\}\Phi(t/T)\,\mathrm{d}t+O(E),\end{aligned} \tag{12}$$

where $\Psi(a+\mathrm{i}t)=\sum a_n n^{-1/2-\mathrm{i}t}$ and $E \ll T^\varepsilon(y^{7/4}+y^{7/8}T^{1/2})$. Here (d,e) and $[d,e]$ denote gcd and lcm. The polynomial-profile extension and finite linear combinations are included in the proof on p. 18. Expanding the two components therefore includes both cross terms, not just their diagonal moments. For our θ, the two error exponents before ε are $1-1.75\cdot10^{-6}$ and $1-0.875\cdot10^{-6}$. A sufficiently small fixed ε absorbs all fixed logarithmic factors and leaves $o(T)$.

The formula is uniform on fixed compact sets of the normalized complex shifts $A=\alpha L$, $B=\beta L$. The apparent poles are combined before differentiating. Cauchy's formula allows the fixed operator $q(-\partial_A)q(-\partial_B)$ to be applied to the main term; the normalized derivatives cancel all powers of L. This proves the bounded derivative moments in Lemma 2 as well. Nonnegative moments pass from smooth weights to the sharp window by upper and lower approximations, taking the height limit before the width of the endpoint smoothing tends to zero. We allow a slightly enlarged fixed height interval for the upper approximation; a finite partition and comparable rescaling reduce it to the published support convention. The fixed positive length margin absorbs these changes.

Here is a finite integral expression for the limiting main term. It specifies precisely the expression to be checked against the arithmetic asymptotic. Label the Conrey component by 0 and the Feng component by 2, and put $F_0=P_1$, $F_2=P_2$. For Mellin variables s,t, set $X=A+s$, $Y=B+t$, $h=s+t$. The four leading kernels are

$$\begin{aligned}K_{00}&=XY/h,\\ K_{20}&=XY/h^3-2Y/h^2+Y/(Xh),\qquad K_{02}=XY/h^3-2X/h^2+X/(Yh),\\ K_{22}&=7XY/h^5-6(X+Y)/h^4+(X/Y+Y/X+8)/h^3\\ &\qquad-2(X^{-1}+Y^{-1})/h^2+(XYh)^{-1}.\end{aligned} \tag{13}$$

In particular the inverse powers of X, Y must not be discarded.

For $0 \le w \le \theta$, define

$$\begin{aligned} D_1(A, w; F) &= F'(w/\theta)/\theta + AF(w/\theta), \\ D_0(A, w; F) &= F(w/\theta), \\ D_{-1}(A, w; F) &= \int_0^w e^{-Az} F((w-z)/\theta)\,\mathrm{d}z. \end{aligned} \tag{14}$$

Extend linearly the following rule on every monomial in (13):

$$cX^e Y^f h^{-n} \longmapsto \frac{c}{\theta^{\ell+m}} \int_0^\theta \frac{v^{n-1}}{(n-1)!} D_e(A, \theta - v; F_\ell) D_f(B, \theta - v; F_m)\,\mathrm{d}v. \tag{15}$$

Its value on $K_{\ell m}$ is denoted $H_{\ell m}(A, B)$. Only $e, f \in \{-1, 0, 1\}$ occur. Define

$$\begin{aligned} I(A, B) &= \sum_{\ell, m \in \{0,2\}} \frac{H_{\ell m}(A, B) - e^{-A-B} H_{\ell m}(-B, -A)}{A + B}, \\ M &= q(-\partial_A) q(-\partial_B) I(A, B)|_{A=B=-R}. \end{aligned} \tag{16}$$

The ordered cross terms are both present. The dual term swaps the two shifts; q differentiates the entire quotient in (16).

Proposition 3. *For (5), the constant in (7) is (16).*

Derivation from the published shifted moment. Insert the coefficient expansion in (12) and use Mellin inversion for each monomial of F_ℓ. The leading zeta ratio has prefactor XY/h. Its logarithmic derivative for one left auxiliary variable is $h^{-1} - X^{-1}$, for one right variable it is $h^{-1} - Y^{-1}$, and for a left–right pair it is h^{-2}. The distinct auxiliary variables represent the copies of Λ. All other connected derivatives for these multiaffine coefficients vanish. Matching k left variables to right variables has multiplicity $\binom{\ell}{k}(m)_k$. Thus the leading factor is

$$\frac{XY}{h} \sum_{k=0}^{\min(\ell,m)} \binom{\ell}{k} (m)_k h^{-2k} (h^{-1} - X^{-1})^{\ell-k} (h^{-1} - Y^{-1})^{m-k}.$$

For $\ell, m \in \{0, 2\}$ this expands to (13). This is the leading-pole extraction in [2, Section 6, (6.3), and pp. 39–40]. The arithmetic Euler factor and its derivatives are treated by that paper's contour argument: after the logarithmic normalization, the omitted terms contribute $O(T/L)$, not a conjectural ratios error. See also the procedure on p. 45. Replacing those factors by one without this error estimate would not suffice.

Use $h^{-n} = \int_0^\infty e^{-hv} v^{n-1}\,\mathrm{d}v/(n-1)!$ in the Mellin inversions. The profiles have support $w \ge 0$ and vanish at zero. Multiplication by $A + s$, by 1, and by $(A + s)^{-1}$ therefore gives respectively the three operators (14), with no endpoint delta term. Their common support restricts v to $[0, \theta]$. The factors $(\log y)^{-\ell-m}$ give $\theta^{-\ell-m}$ after rescaling, proving (15). The second term of (12) gives the dual shift $(-B, -A)$ and the factor e^{-A-B}, with the minus sign from the pole of $\zeta(1 - \alpha - \beta)$. This gives (16); the contour estimates and shift uniformity above justify applying $q(-\partial_A) q(-\partial_B)$. □

As a normalization check, deleting the Feng component reduces (16) to

$$1 + \theta^{-1} \int_0^1 \int_0^1 e^{2Rv} [P_1'(x) q(v) + \theta P_1(x)(Rq(v) + q'(v))]^2\,\mathrm{d}x\,\mathrm{d}v.$$

This check is not a substitute for the arithmetic identification in Proposition 3.

4 Finite certificate and completion

Lemma 4. *For the rational data (1)–(4), the functional (16) satisfies the outward enclosure*

$$1.12818163807203565629105 < M < 1.12818163807203565629124. \tag{17}$$

Moreover,

$$1.12820541062467513417945 < e^{R(1-0.8845)} < 1.12820541062467513417947. \tag{18}$$

Finite verification and its error bound. The attached interval evaluator computes (15) without quadrature. It forms normalized Taylor jets in both shifts through order six. In D_{-1}, expand e^{-Az} through degree $N = 24$. For $C_F = \sum_d |F_d|$, the error in its ath normalized shift derivative at $A = \pm R$ is bounded by

$$\frac{3C_F\theta^{1+a}}{a!(1+a)}\frac{(R\theta)^{N+1}}{(N+1)!}, \qquad 0 \le a \le 6, \tag{19}$$

because $R\theta < 1$ and $e^{R\theta} < 3$. The corresponding full-derivative bound is the first factor in (19). The finite polynomial integrals are evaluated by

$$\int_0^w z^j(w-z)^d\,\mathrm{d}z = \frac{j!d!}{(j+d+1)!}w^{j+d+1}, \qquad \int_0^\theta \frac{v^{n-1}(\theta-v)^d}{(n-1)!}\,\mathrm{d}v = \frac{d!}{(n+d)!}\theta^{n+d}.$$

For a product of two truncated factors, combine the two errors with their full-factor bounds and the product of errors; contract with absolute values of the rational kernel and operator coefficients. The summed analytic error is less than $8.799\cdot 10^{-23}$. All remaining operations, including the dual exponential e^{2R}, use outward-rounded `mpmath.iv` intervals at 55 decimal digits. They yield a narrower interval than (17). An independent 40-digit, degree-20 setting also passes the threshold. The wrapper `verify.py` checks the stated enclosures, exact polynomial constraints, and the four expanded kernels. This is a numerical certificate using a specified interval library, not a formal verification of that library or of Proposition 3. □

Combining Lemma 2, Proposition 3, and Lemma 4 proves the proposed Theorem 1. The strict gap in (17)–(18) is greater than $2.37725\cdot 10^{-5}$ in moment units. Thus the conclusion uses a fixed rational witness, not convergence or global optimality of a numerical optimizer. Direct high-precision evaluation gives $1 - \log M/R = 0.884520175542196\ldots$, but the claimed bound is 0.8845.

Verification priorities for the referee

Review should prioritize the phase-counting convention in (11), the shift-uniform applicability of (12) to (5), and Proposition 3. The code establishes only Lemma 4. The separate audit guide gives reproduction commands, review questions and the literature-search scope.

References

[1] X. Wu, *Distinct zeros of the Riemann zeta-function*, arXiv:1206.3737v2 (2012), especially Section 3, pp. 6–8. https://arxiv.org/pdf/1206.3737v2.

[2] K. Pratt, N. Robles, A. Zaharescu and D. Zeindler, *More than five-twelfths of the zeros of ζ are on the critical line*, arXiv:1802.10521v3, Theorem 4.1 and Sections 6–7. https://arxiv.org/pdf/1802.10521v3.

[3] L. Alpöge and R. Furman, *More than two thirds of the zeta zeros are simple and on the critical line*, arXiv:2608.13637v2 (2026), Remark 7.1. https://arxiv.org/html/2608.13637v2#S7.SS3.